

DeepTutor AI — Complete API Documentation

Generated on August 07, 2026 • Architecture & Endpoint Guide

1. User Authentication (/api/auth)

ENDPOINT	METHOD	DESCRIPTION & PURPOSE
/api/auth/register	POST	Create Account: Registers a new student with an email, username, and password.
/api/auth/login	POST	Sign In: Checks credentials and returns a secure JWT bearer token.
/api/auth/me	GET	Who Am I?: Fetches current authenticated user profile & role.

2. AI Chat & SSE Streaming (/api/chat)

ENDPOINT	METHOD	DESCRIPTION & PURPOSE
/api/chat/sessions	POST	Start New Chat: Creates a new chat session for a study topic or PDF.
/api/chat/sessions	GET	List Chats: Shows all your past chat conversations sorted by date.
/api/chat/sessions/{id}/messages	GET	Chat History: Fetches all messages in a specific chat session.
/api/chat/sessions/{id}/message	POST	Send Message: Sends a question to the AI tutor and gets a response.
/api/chat/sessions/{id}/message/stream	GET	Live SSE Stream: Streams AI answers token-by-token with page sources [p.X] & knowledge graph context.
/api/chat/sessions/{id}	DELETE	Delete Chat: Removes a chat session and its history.

3. Documents & Knowledge Graph Vectorization (/api/documents)

ENDPOINT	METHOD	DESCRIPTION & PURPOSE
/api/documents/upload	POST	Upload Study Material: Parses text, vectorizes chunks to ChromaDB, and extracts NetworkX knowledge graph.
/api/documents/topic/{topic_id}	GET	Get My Files: Shows all uploaded documents for a study section & indexing status.
/api/documents/{doc_id}/status	GET	Check Indexing: Polls whether document processing is pending, 50%, or done.
/api/documents/topic/{topic_id}/graph	GET	Get Knowledge Graph: Returns 3D nodes and relationships extracted from your PDF.

4. Quiz Engine (/api/quiz)

ENDPOINT	METHOD	DESCRIPTION & PURPOSE
/api/quiz/suggestions	GET	Topic Suggestions: Extracts key concepts from PDF to offer clickable quiz topic pills.
/api/quiz/generate	POST	Generate Quiz: Creates a multi-choice quiz based on topic, difficulty, and question count (3-20 Qs).

/api/quiz/{id}	GET	Fetch Quiz: Gets questions and options for a created quiz.
/api/quiz/{id}/submit	POST	Submit Answers: Grades quiz, calculates score %, awards XP points, and saves attempt.
/api/quiz/my-attempts	GET	Quiz History: Shows all past quiz scores and attempts.

5. Flashcards (/api/flashcards)

ENDPOINT	METHOD	DESCRIPTION & PURPOSE
/api/flashcards/generate	POST	Generate Deck: Creates Q&A; study flashcards directly from PDF document pages.
/api/flashcards/topic/{topic_id}	GET	List Flashcards: Fetches all flashcards saved under a study topic.
/api/flashcards/{id}/review	POST	Mark Mastered: Toggles whether you have mastered a flashcard.

6. AI Study Roadmap (/api/study-plan)

ENDPOINT	METHOD	DESCRIPTION & PURPOSE
/api/study-plan/generate	POST	Create Study Plan: Generates a day-by-day study roadmap based on target date.
/api/study-plan/topic/{topic_id}	GET	Get Current Plan: Returns study roadmap schedule and completed days.
/api/study-plan/{id}/complete-day	POST	Check Off Day: Marks Day 1, Day 2, etc. as completed and awards XP.

7. Gamified XP & Progress (/api/progress)

ENDPOINT	METHOD	DESCRIPTION & PURPOSE
/api/progress/summary	GET	Dashboard Metrics: Returns total XP, Level, active streak, quiz avg, and flashcards mastered.
/api/progress/calendar	GET	Activity Heatmap: Returns daily activity counts for study calendar heatmap.
/api/progress/report	GET	Download Report: Generates a downloadable text summary report of achievements.

8. Leaderboard (/api/leaderboard)

ENDPOINT	METHOD	DESCRIPTION & PURPOSE
/api/leaderboard/top	GET	Global Leaderboard: Ranks top learners by total XP, level, and streak.

9. Model Context Protocol (/api/mcp)

ENDPOINT	METHOD	DESCRIPTION & PURPOSE
/api/mcp/tools	GET	List AI Tools: Exposes available tools for external AI agent integrations.
/api/mcp/call	POST	Execute Tool: Runs tool calls (e.g. vector search) and returns JSON context.